How the HY Analyst works, step by step

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The whole method in one line. Measure the issuer today, project it under each macro story, price what the bond should pay, and buy only where the market is wrong by more than the cost.

Three phases, eleven steps. Measure (steps 1 to 4) → Forecast (steps 5 to 7) → Decide (steps 8 to 11).

One rule over all of it. Every number carries the date the market could first read it. No step reads a number from the future.

Phase 1: Measure the issuer today

1. Load the data with its release time

Do	Pull financials (Compustat point-in-time, SEC filings), bond prices (TRACE), and the Treasury curve. Stamp each number with `available_at`.
Why	A backtest that reads a number before its release finds profit that never existed.
You get	Quarterly records, every version of every quarter.
Where	Methodology §3; `hy/inputs.py`, `hy/sources/`

2. Frame the business

Do	Pick one archetype (for example lease-heavy, cyclical, asset-heavy contracted) and up to two modifiers (sponsor, decline, concentration).
Why	The archetype sets which EBITDA to use, how the business fails, and which events to watch.
You get	`archetype`, `modifiers`, a one-line reason.
Where	Framework F2

3. Measure health now

Do	Adjust EBITDA for one-off items. Compute free cash flow, liquidity, leverage, coverage, covenant headroom, risk flags, and the rating gap.
Why	These are the inputs every later step uses.
You get	One state vector per issuer per filing.
Where	Methodology §4; `hy/state.py`

4. Check the two clocks

Do	Put fixed charges (rent, debt, interest) and signed revenue on one yearly grid.
Why	When the bills outlast the contracts, the bond is a bet on renewal.
You get	Breakeven margin, renewal-dependent charges, largest shortfall.
Where	Methodology §4.7; `hy/clocks.py`

Phase 2: Forecast under each macro story

5. Turn forecasts into macro paths

Do Read GDP, inflation, rate, and spread forecasts with their vintages. Build five paths: soft landing, recession, higher for longer, disinflation rally, stagflation. Weight each path by the forecasts, then update with the cycle phase.
Why The forecasters' own uncertainty sets the odds, not the analyst's mood.
You get Five paths of eight quarters, each with a probability; prior, forecast mass, and posterior recorded.
Where Methodology §5.4; `hy/forecasts.py`, `hy/macro.py`

6. Run every case under every path

Do Build six cases (management, consensus, base, moderate downside, severe downside, upside). Run each one under each path. Refinance each maturity at the forward Treasury plus fair spread, or record a failure.
Why One case on one path hides the tail; the grid shows it.
You get 30 projected paths of revenue, EBITDA, debt, cash, and headroom.
Where Methodology §5.1; framework F6; `hy/forward.py`

7. Weight the cases and average

Do Weight the cases from the F6 table (later the fitted logistic). If the downside weight passes 0.5, shift every weight down one step (F6.6). Average across cases and paths.
Why The expected number must include how likely each story is.
You get Expected leverage and its spread, breach, liquidity, and refinance odds, per path and weighted.
Where Methodology §5.3; framework F6.6

Phase 3: Decide what to own

8. Price what the bond should pay

Do Spread from expected loss (default probability times loss), times the market price of risk, plus a model for the rest. Compare with the market spread. Flag cheap only outside the error band.
Why A bond is cheap against everything the model knows, not against a noisy point.
You get Fair spread, residual, cheap or rich flag.
Where Methodology §6, §7; `hy/recovery.py`, `hy/fairspread.py`

9. Convert to expected alpha

Do Per path: carry less the hedge, plus the part of the gap that closes, less expected default loss. Weight by path, add event P&L, subtract trading cost.
Why The number you act on is return after risk and cost, not spread.
You get `exp_alpha`, its parts, its spread across paths, and the information ratio.
Where Methodology §7.5; `hy/alpha.py`

10. Signal, gate, and rank

Do Score nine signals. Pass the gate: a gap larger than its noise, a dated catalyst, alpha above cost, one observable that would prove the call wrong. Rank the whole universe and name the position each new one replaces.
Why A credit opinion is not a trade until the price is wrong by enough.
You get The force rank, with `replaces_position` and `invalidation` for each entry.
Where Methodology §8; framework F9, F10; `hy/signals.py`

11. Size for events, then check the record

Do For each dated event, list outcomes and their odds under each path; compute P&L per outcome; cap size to the event budget. After each event and each exit, record what happened against what was predicted.
Why The model that calls a bond cheap must also say how much it loses on a miss, and must be scored.
You get Event matrix, limits, hedges, postmortem.
Where Methodology §9 to §11; framework F11, F13; `hy/events.py`

What you supply, and what the system computes

- **You:** the archetype, analyst cases and gaps to consensus, ratings, document terms, and any override in `data/analyst_inputs/`.
- **The system:** everything else, each number with its source and release time.
- **Every hand-set number** has a decision page in `docs/decisions/` with the test that would change it.